
Nisaba is an embedded-first, zero-dependency, ultra-lightweight 2D graphics rendering and layout engine written from scratch in Modern C++ (C++20) â Visit <https://github.com/vaxp/nisaba> for updates.

Note

Nisaba delivers radical technological sovereignty with zero external dependencies and 100% pure ISO C++20 compliance.

Tip

Press Ctrl + Shift + P to open Command Palette, or use formula $E = mc^2$ and chemistry H2O with Hardware Acceleration.

Important

Multi-backend rendering supports native Vulkan 1.0+ and OpenGL ES 3.2 with 4x hardware MSAA and anti-banding dithering.

Warning

Embedded DRM/KMS dumb buffer targets require zero-copy pitch alignment with `PixmapMut::from_raw_parts`.

Caution

Do not link third-party windowing libraries (GLFW/SDL). Use Nisaba's native sovereign `backend_os` subsystem.

â... Sovereign Hardware GPU & Codec Architecture (Click to Toggle)

The GPU pipeline leverages direct SPIR-V bytecode shaders with hardware 4x MSAA resolve. Sovereign codecs encode HD QOI in under 7.9 ms ð ¥

- High-precision subpixel anti-aliasing (256 coverage levels)
- Hardware accelerated GPU rendering (Vulkan 1.0+ & OpenGL ES 3.2)
- Sovereign image codecs: PNG, JPEG, QOI, Deflate (RFC 1950/1951)
- Sovereign typography with TrueType parsing, BiDi (UAX #9) & Arabic shaping
- GitHub-Grade Markdown Engine with HTML tags and Callout Alerts ð
- Sovereign ISO 32000 PDF document engine

Operation Category	Nisaba CPU (µs)	Cairo 1.18 (µs)	Speedup Ratio
Diagonal Full-HD Gradient	308.7 µs	6653.0 µs	21.55x (Nisaba)
2-Point Focal Spotlight	327.1 µs	5231.8 µs	15.99x (Nisaba)
Radial Glow Gradient	288.6 µs	4000.8 µs	13.86x (Nisaba)
25 Concentric Rings	461.4 µs	6002.2 µs	13.01x (Nisaba)
Dashed Orbit Ring	31.7 µs	364.1 µs	11.48x (Nisaba)
25 Alpha Circles Blend	68.4 µs	703.0 µs	10.27x (Nisaba)
TOTAL 25-SUITE FRAME	5972.5 µs	44968.7 µs	7.53x Faster

cpp

```
#include <nisaba/nisaba.hpp>
using namespace nisaba;

int main() {
    // Allocate a high-definition 2D canvas
    auto doc = markdown::MarkdownDocument::from_string(content);
    auto pixmap = Pixmap::create(1080, 800);
    Canvas canvas(*pixmap);
    markdown::MarkdownRenderer::render(canvas, doc, 40.0f, 20.0f, 1);
    pixmap->save_image("output.png");
    return 0;
}
```

[1] Verified via 13/13 automated test suites in Nisaba test runner. â'